Analysis Tutorial 10

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1 Introduction

In this tutorial, we will review previous topics from the course and we will also study **series of real numbers** and the concept of **continuity of functions**. We will recall definitions, convergence criteria, and then solve problems that develop intuition and proof skills for these fundamental analytical concepts.

2 Cartesian Product of Sets

Definition. For sets A, B, the Cartesian product is

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

Problems

1. Let $A = \{1, 2, 3\}$, $B = \{x, y\}$. Define a relation $R \subseteq A \times B$ such that every $a \in A$ is related to exactly one $b \in B$.
   (a) How many such relations exist?
   (b) How many are functions from A to B?
2. Let A, B, C be sets. Prove that

   $$(A \times B) \times C \neq A \times (B \times C)$$

   as sets, but that they are *naturally isomorphic*. Define this bijection explicitly.

3 Relations and Special Types

Definition. A relation $R \subseteq A \times A$ can be:

- Reflexive: $(a, a) \in R$ for all a.
- Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$.
- Transitive: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.

A **partial order** is reflexive, antisymmetric, transitive. An **equivalence relation** is reflexive, symmetric, transitive.

Problems

1. Let $R \subseteq \mathbb{Z} \times \mathbb{Z}$ be defined by $aRb \iff a - b$ is divisible by 3.
   (a) Prove R is an equivalence relation.
   (b) Find all equivalence classes.

4 Equivalence Relations and Partitions

Theorem. Every equivalence relation R on a set A partitions A into disjoint equivalence classes.

Proof Sketch. Define $[a] = \{x \in A \mid xRa\}$.

- $a \in [a]$ (non-empty),
- Either $[a] = [b]$ or $[a] \cap [b] = \varnothing$,
- $\bigcup_{a \in A} [a] = A$.

5 Natural Numbers and Peano Axioms

Peano Axioms:

1. 0 is a natural number.
2. Every natural number n has a successor $S(n)$.
3. $S(n) \neq 0$ for all n.
4. $S(n) = S(m) \Rightarrow n = m$.
5. (Induction) If $0 \in X$ and $n \in X \Rightarrow S(n) \in X$, then $X = \mathbb{N}$.

Problems

1. Define addition recursively:
$$a + 0 = a, \quad a + S(b) = S(a + b).$$
Prove that addition is associative.
2. Define multiplication recursively:
$$a \cdot 0 = 0, \quad a \cdot S(b) = a \cdot b + a.$$
Prove distributivity: $a(b + c) = ab + ac$.

6 Bijection, Injection, Surjection

Definition. A function $f : A \to B$ is:

- Injective if $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$.
- Surjective if for every $b \in B$ there exists $a \in A$ with $f(a) = b$.
- Bijective if both injective and surjective.

7 Construction of Integers

Define $\sim$ on $\mathbb{N} \times \mathbb{N}$ by
$$(a, b) \sim (c, d) \iff a + d = b + c.$$
Each equivalence class represents an integer $[(a, b)]$.

Define operations:
$$[(a, b)] + [(c, d)] = [(a + c, b + d)], \quad [(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)].$$

Problems

1. Prove $\sim$ is an equivalence relation.

8 Construction of Rationals

Define $\sim$ on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ by

$$(a, b) \sim (c, d) \iff ad = bc.$$

Equivalence classes are the rational numbers.

9 Construction of Real Numbers via Cauchy Sequences

Definitions.

- A **Cauchy sequence** (x_n) in $\mathbb{Q}$ satisfies: $\forall \epsilon > 0, \exists N, \forall m, n > N, |x_n - x_m| < \epsilon$.
- A **null sequence** satisfies $\lim x_n = 0$.
- Define $(x_n) \sim (y_n)$ if $(x_n - y_n)$ is null.

Then $\mathbb{R}$ is the set of equivalence classes of Cauchy sequences of rationals.

10 Archimedean Property and Density

Archimedean Property. For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > x$.

Density. Between any $x < y$ in $\mathbb{R}$, there exists $r \in \mathbb{Q}$ with $x < r < y$.

Part I: Series of Real Numbers

Definition (Series and Partial Sums)

Let (a_n) be a sequence of real numbers. The **series** $\sum_{n=1}^{\infty} a_n$ is defined as the limit of the sequence of partial sums

$$s_N = \sum_{n=1}^{N} a_n.$$

If $\lim_{N \to \infty} s_N = S$ exists and is finite, we say the series *converges* and write

$$\sum_{n=1}^{\infty} a_n = S.$$

Otherwise, the series *diverges*.

Definition (Absolute and Conditional Convergence)

A series $\sum a_n$ is said to converge **absolutely** if $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, we say it converges **conditionally**.

Definition (Geometric Series)

The geometric series is

$$\sum_{n=0}^{\infty} ar^n,$$

which converges if and only if $|r| < 1$, and in that case

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}.$$

Problems

Problem 1

Prove that if $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$. Give an example of a sequence (a_n) such that $\lim a_n = 0$ but $\sum a_n$ diverges.

Problem 2

Prove that if $\sum a_n$ converges, then for any $\epsilon > 0$, there exists N such that for all $m > n > N$,

$$\left| \sum_{k=n+1}^{m} a_k \right| < \epsilon.$$

(This is the Cauchy criterion for convergence of series.)

Part II: Continuity of Functions

Definition (Continuity at a Point)

Let $f : \mathbb{R} \to \mathbb{R}$. We say that f is **continuous at** $x_0 \in \mathbb{R}$ if

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ such that } |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Equivalent Definition (Limit Form)

A function f is continuous at x_0 if and only if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Properties of Continuous Functions

If f and g are continuous at x_0, then:

$$f + g, \quad f - g, \quad fg, \quad \frac{f}{g}\ (g(x_0) \neq 0)$$

are also continuous at x_0.

Problems

Problem 1

Let

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Prove that f is continuous at $x = 0$.

Problem 2

Prove that if f and g are continuous at x_0, then so are $f + g$ and $f \cdot g$.